UniGear Connect

Software Requirements Specification & Conceptual Model

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Prepared for the System Modeling course project. This document defines the system requirements, conceptual analysis model, UML diagrams, business rules, data expectations, and traceability views for the UniGear Connect university equipment borrowing platform.

Version 1.0

1. Introduction

UniGear Connect is a university equipment borrowing system created to digitize and organize the lending of academic resources such as laptops, cameras, projectors, VR headsets, and similar lab assets. The system is intended to replace manual paper logs and disconnected spreadsheets with a structured, role-based platform that supports visibility, accountability, and operational efficiency.

The problem domain is practical and common in universities: students need fair access to limited equipment, lab managers need a reliable method for approving requests and tracking returns, and administrators need control over users, rules, and system settings. In the current manual process, availability is not visible in real time, double booking is possible, overdue items are difficult to trace, and disputes over item condition or responsibility are harder to resolve.

The purpose of this document is to define the functional and non-functional requirements of UniGear Connect and to present the conceptual analysis model that explains how the domain works. The document follows the D2 course expectations by combining a Software Requirements Specification with a Conceptual Model supported by UML diagrams and traceability tables.

1.1 Scope and System Overview

The system serves three primary user roles. A Student acts as the borrower who searches for equipment, creates reservation requests, views active loans, and receives reminders. A Lab Manager supervises operational activities such as approving or rejecting reservations, performing check-out and check-in operations, updating equipment status, and generating overdue reports. A System Administrator manages users, controls borrowing rules, and maintains the overall configuration of the platform.

The system boundary includes user authentication with university identity data, equipment catalog browsing, reservation workflow, check-in and check-out support, automatic due-date reminders, display-only penalty calculation, and administrative inventory management. Online payment processing, procurement of new equipment, advanced maintenance ticketing, and inter-campus logistics remain outside scope. This boundary is intentionally narrow enough to keep the solution realistic and modelable while still addressing a meaningful operational problem.

1.2 Stakeholders and Intended Benefits

- Students benefit from transparent availability information, fairer access to shared resources, and visibilityinto due dates and reservation outcomes.
- Lab Managers benefit from a structured approval workflow, faster check-out/check-in processing, clearerinventory status, and overdue monitoring.
- System Administrators benefit from centralized user control, configurable borrowing rules, and a consistentfoundation for future maintenance and reporting.
- University administration benefits from improved accountability, lower loss risk, reduced manual effort, andbetter utilization of academic equipment.

2. Domain Explanation and Conceptual Narrative

The domain of UniGear Connect revolves around controlled access to shared physical resources. Equipment is the core asset of the domain. Each equipment item belongs to a category, has a current status, and can participate in multiple reservations and loans over time. A reservation represents a request to borrow an item for a planned time range. A loan represents the confirmed operational borrowing event that begins at check-out and ends at return.

A reservation does not automatically mean that equipment has been borrowed. First, availability must be checked and the request must be reviewed by the lab manager. Once approved, the item can be checked out to the student, which creates the corresponding loan record. If the item is returned late, the system determines the delay and produces a penalty record. Notifications support communication by reminding students about due dates and system events. Reports aggregate operational data for managerial oversight, especially for overdue items and inventory monitoring.

This conceptual separation between reservation, loan, equipment state, and penalty is important because it reflects the real-world process rather than implementation convenience. The domain therefore supports both planned borrowing and actual borrowing, including exceptions such as rejection, overlap prevention, unavailable items, overdue returns, and temporarily non-borrowable assets due to maintenance.

From an analysis perspective, the domain is driven by a lifecycle: equipment is searched, reserved, approved or rejected, checked out as a loan, and later returned. Some objects, such as penalties and reports, arise only under certain operational conditions. This lifecycle-oriented understanding is what guides the conceptual model and prevents the design from collapsing different real-world meanings into one generic record type.

3. Functional Requirements

The following functional requirements define what UniGear Connect must do. Each statement is written in a testable “shall” form and is aligned with the project scope established in the earlier system brief.

FR-01 — The system shall allow students to search the equipment catalog using category, availability, and keyword filters.

This requirement supports fast discovery of borrowable resources and reduces time lost in manual inquiry.

FR-02 — The system shall allow students to view equipment details including category, current status, and relevant specifications before creating a reservation.

Students need enough contextual information to decide whether a resource is appropriate for their academic task.

FR-03 — The system shall allow students to submit a reservation request for a selected equipment item and a defined borrowing period.

A formal request mechanism is necessary to create traceable and reviewable borrowing intent.

FR-04 — The system shall validate availability before storing a reservation request and shall prevent overlapping reservations for the same equipment item.

This enforces fair allocation and directly addresses the double-booking problem described in D1.

FR-05 — The system shall allow lab managers to approve or reject reservation requests and shall record the resulting reservation status.

The approval workflow ensures human oversight for limited or sensitive equipment.

FR-06 — The system shall support check-out and check-in operations using QR code scanning or manual ID entry.

Operational handling must be quick, practical, and suitable for real lab environments.

FR-07 — The system shall allow students to view their active loans, due dates, and reservation outcomes.

Borrowers need visibility into their responsibilities and request status.

FR-08 — The system shall allow lab managers to update equipment status to Available, Reserved, Borrowed, or Maintenance.

Equipment state management is required for accurate availability and operational control.

FR-09 — The system shall automatically send reminder notifications before due dates through the university email service or in-app messaging.

Timely reminders reduce late returns and improve compliance.

FR-10 — The system shall calculate and display overdue penalties when an item is returned after its due date.

The project scope includes penalty logic and visibility even though payment processing is out of scope.

FR-11 — The system shall record borrowing history for each equipment item and each student.

History supports accountability, reporting, and future auditing.

FR-12 — The system shall allow system administrators to manage user accounts and configure borrowing rules such as maximum concurrent loans or permitted borrowing duration.

Administrative control is needed to align the platform with institutional policy.

4. Use Cases

The selected use cases cover the main interactions expected from the system. Together they include the required mixture of creation, state change, report generation, exception handling, and business-rule enforcement specified in the course project requirements.

UC-01 Search Equipment

Goal: Allow a student to locate suitable equipment based on need and availability.

Primary Actor: Student

Preconditions: The student is authenticated.

Trigger: The student opens the equipment catalog or enters a search query.

Main Success Scenario:

1. The student selects one or more filters such as category, availability, or keyword.
2. The system validates the search input and retrieves matching equipment records.
3. The system displays a list of equipment items with summary information such as item name, category, andcurrent status.
4. The student may open an item to view additional details.

Alternate / Exception Flows:

- If no equipment matches the criteria, the system displays an empty-result message and allows the studentto refine the filters.
- If the catalog cannot be loaded temporarily, the system informs the user and logs the issue.

Postconditions: Relevant equipment information is displayed or an informative no-result/temporary-error message is shown.

UC-02 Create Reservation

Goal: Allow a student to request a borrowing period for a selected item.

Primary Actor: Student

Preconditions: The student is authenticated and has permission to borrow equipment.

Trigger: The student selects an equipment item and chooses to reserve it.

Main Success Scenario:

1. The system displays the selected item and a reservation form.
2. The student enters the intended borrowing dates.
3. The system checks policy constraints such as borrowing limit and requested date validity.
4. The system checks whether the equipment is available for the requested interval.
5. If all validations pass, the system creates a reservation with Pending status.
6. The system confirms that the reservation request has been submitted for review.

Alternate / Exception Flows:

- If the selected dates overlap with an existing reservation or loan, the system rejects the request andexplains the conflict.
- If the student exceeds the borrowing limit, the system blocks submission and displays the relevant rule.
- If the item is under Maintenance, the system does not allow reservation creation.

- **Postconditions:** A new pending reservation exists, or the student receives a clear explanation of why the request was not accepted.

UC-03 Review Reservation Request

Goal: Allow a lab manager to approve or reject a pending request.

Primary Actor: Lab Manager

Preconditions: The lab manager is authenticated and at least one pending reservation exists.

Trigger: The lab manager opens the pending reservations list.

Main Success Scenario:

1. The system displays pending requests with student, equipment, and date information.
2. The lab manager selects one request for review.
3. The system shows the request details and current availability context.
4. The lab manager chooses to approve or reject the request.
5. The system records the decision and updates the reservation status.
6. The system notifies the student about the outcome.

Alternate / Exception Flows:

- If the equipment becomes unavailable before approval is completed, the system prevents approval andrequests manager re-evaluation.
- If the selected reservation was already processed by another manager, the system refreshes the recordand displays the new status.

Postconditions: The selected reservation is marked Approved or Rejected and the student is informed.

UC-04 Check-Out Equipment

Goal: Create an operational loan when approved equipment is handed to a student.

Primary Actor: Lab Manager

Preconditions: The lab manager is authenticated, the reservation is approved, and the equipment is physically available.

Trigger: The student arrives to collect the reserved equipment.

Main Success Scenario:

1. The lab manager scans the equipment QR code or enters the equipment ID manually.
2. The system retrieves the approved reservation and borrower information.
3. The lab manager verifies the item and borrower.
4. The system creates a loan record with borrow date and due date.
5. The system changes the equipment status to Borrowed.
6. The system confirms successful check-out.

Alternate / Exception Flows:

- If no approved reservation exists, the system blocks check-out.
- If the item is already marked Borrowed or Maintenance, the system blocks the operation and shows thecurrent status.

Postconditions: A valid loan record is created and the equipment is marked Borrowed.

UC-05 Return Equipment

Goal: Complete the borrowing lifecycle and restore availability when an item is returned.

Primary Actor: Lab Manager

Preconditions: The lab manager is authenticated and an active loan exists for the item.

Trigger: Returned equipment is presented for check-in.

Main Success Scenario:

1. The lab manager scans the equipment QR code or enters the ID.
2. The system retrieves the active loan and associated due date.
3. The lab manager confirms the return.
4. The system records the return date and changes equipment status to Available unless another exceptionalstate applies.
5. The system checks whether the return is late.
6. If late, the system generates and displays a penalty entry.
7. The system confirms successful check-in.

Alternate / Exception Flows:

- If no active loan is found for the item, the system does not complete check-in and informs the lab manager.
- If the item is returned damaged and the process requires manual follow-up, the system may log the returnwhile flagging the item for further review or Maintenance.

Postconditions: The loan is closed, the equipment state is updated, and any overdue penalty is recorded.

UC-06 Generate Overdue Report

Goal: Provide a summary of current overdue items for managerial oversight.

Primary Actor: Lab Manager

Preconditions: The lab manager is authenticated and has reporting permission.

Trigger: The lab manager requests an overdue report.

Main Success Scenario:

1. The system identifies active loans whose due date has passed.
2. The system compiles report data including borrower, equipment, due date, and delayed duration.
3. The system presents the report on screen and stores report metadata if configured.
4. The lab manager reviews the report for follow-up actions.

Alternate / Exception Flows:

- If no overdue items exist, the system produces an empty report with a clear message.
- If reporting data cannot be generated because of a temporary technical issue, the system reports theproblem and logs it.

Postconditions: An overdue report is produced or a justified empty/temporary-error result is shown.

5. Non-Functional Requirements

The non-functional requirements define the quality expectations of the system. They are especially important because the project is intended to model a realistic operational platform rather than a minimal prototype.

NFR-01 Security — The system shall require authentication through university identity data and shall enforce role-based authorization so that students, lab managers, and administrators can access only the functions permitted to their roles.

Security is critical because the system manages university assets and user data.

NFR-02 Performance — For standard operations such as search, request submission, active loan viewing, and reservation review, the system shall provide a response within 2 seconds under normal academic usage conditions.

The KPI target in D1 emphasizes fast operation and reduced process time.

NFR-03 Reliability — The system shall preserve consistent reservation, loan, and status data even if two users attempt conflicting actions close in time, and it shall prevent partial updates that leave objects in invalid states.

This requirement protects operational trust in the platform.

NFR-04 Usability — The interface shall present the core actions for each role clearly enough that a first-time user can complete frequent tasks such as searching or approving a reservation without training beyond a short orientation.

Low friction is necessary because staff resistance to change was identified as a project risk.

NFR-05 Maintainability — The system shall separate user management, reservation workflow, inventory operations, and notification logic into clearly organized modules so future policy or interface changes can be applied with limited ripple effects.

The course project expects a system that could guide implementation teams later on.

NFR-06 Privacy and Auditability — The system shall store only necessary user data, maintain activity logs for critical actions such as approval, rejection, and status updates, and preserve records according to university policy.

This balances accountability with responsible data handling.

6. Data Requirements

The system manages several key data objects. At this stage the purpose is not to define implementation schema details, but to clarify what information the system must retain in order to support the required behavior.

Data Object	High-Level Stored Data
User	Store unique user ID, name, email, role, and role-specific information required for authorization and communication.
Equipment	Store equipment ID, item name, category, status, and basic descriptive details needed for search and operational identification.
Reservation	Store reservation ID, requester, equipment reference, start date, end date, status, and creation timestamp.
Loan	Store loan ID, linked reservation if applicable, borrower, equipment reference, borrow date, due date, and return date.
Data Object	**High-Level Stored Data**
Penalty	Store penalty ID, linked loan, amount, and number of delayed days.
Notification	Store notification ID, recipient, message, date/time, and read state.
Report	Store generated report metadata such as report ID, generation date, type, and generating manager when reports are retained.

Data integrity assumptions include prevention of contradictory equipment states, protection against overlapping reservations, and preservation of historical records for accountability. Privacy assumptions include storing only institutionally necessary user information and limiting visibility of administrative actions to authorized roles.

7. Acceptance Criteria / Test Conditions

The following concise test conditions provide a direct basis for validating both the requirements and the user-facing behavior of the system.

ID	Acceptance Condition
AC-01	Given a logged-in student on the search page, when the student filters by category and availability, then the system returns only matching equipment items within 2 seconds.
AC-02	Given an equipment item already reserved for an overlapping time interval, when a student attempts to reserve the same item for the overlapping period, then the system rejects the request and explains that the item is unavailable.
AC-03	Given a pending reservation request, when a lab manager approves it, then the reservation status changes to Approved and becomes eligible for check-out.
AC-04	Given an approved reservation, when the lab manager performs check-out by scanning the item or entering the ID, then the system creates a loan record and changes the equipment status to Borrowed.
AC-05	Given an active loan that is returned after its due date, when check-in is completed, then the system computes delay days and displays the corresponding penalty.
AC-06	Given a student with active loans, when the student opens the personal dashboard, then the system shows all active loans and their due dates.
AC-07	Given a due date approaching according to system rules, when the reminder threshold is reached, then the system sends a reminder notification to the student.
AC-08	Given an administrator updates the maximum concurrent loan policy, when a student later exceeds that limit, then the system blocks the new reservation request.

8. Business Rules

Business rules express core domain constraints and policies that the software must enforce consistently. They are not merely interface behaviors; they reflect the institutional rules of borrowing and resource control.

BR-01 — A reservation may be approved only if the requested equipment is available for the selected period and is not marked as Maintenance or Borrowed.

BR-02 — A student may not hold more active loans or approved reservations than the configured borrowing limit.

BR-03 — A loan can be created only from an approved reservation during a valid check-out operation.

BR-04 — If the current date exceeds the due date at return time, the system shall compute the overdue duration and generate a corresponding penalty entry.

9. UML Use Case Diagram

The Use Case Diagram provides the external behavioral view of the system. It identifies who interacts with UniGear Connect and which major services each actor may initiate. This view is useful because it makes the system boundary explicit and keeps the requirements anchored in observable behavior rather than internal implementation details.

Three design choices are worth noting. First, student actions are separated from managerial and administrative actions to reflect clear role responsibilities. Second, the reservation workflow is not modeled as a single generic action; instead, creation and review are represented separately because they occur at different moments and are executed by different actors. Third, the diagram includes an availability check as a supporting behavior for reservation creation, emphasizing that request submission depends on rule validation.

- Student-side use cases focus on discovering resources, creating requests, and observing loan-relatedinformation.
- Lab-manager use cases focus on operational control, especially approval decisions, check-out/check-in,inventory state updates, and reporting.
- System-administrator use cases remain limited to user and policy management in order to respect theproject scope.

Figure 1. Use Case Diagram

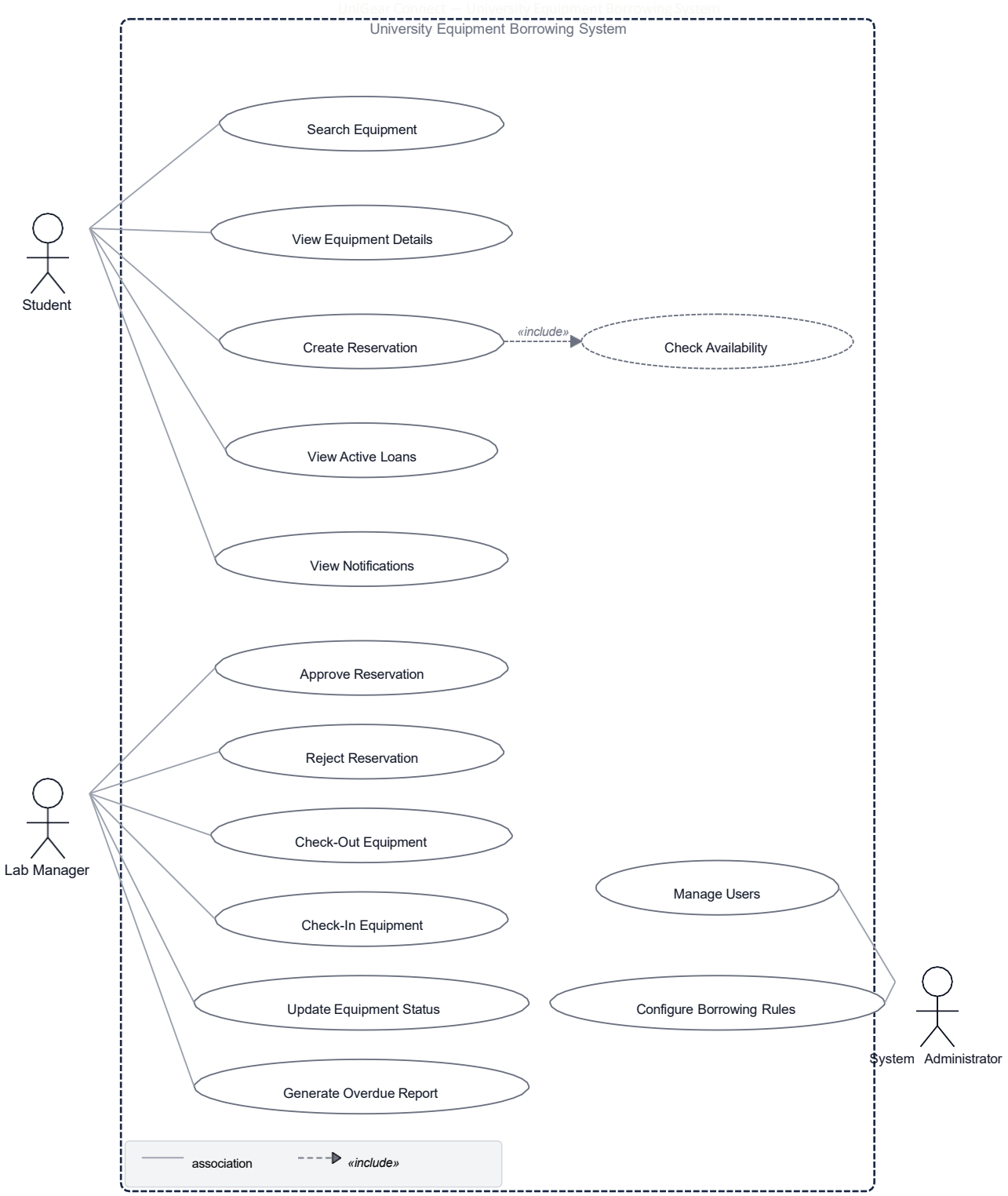


Figure 1. Use Case Diagram for UniGear Connect.

10. UML Activity Diagram

The Activity Diagram models the flow of the Create Reservation process from the moment a student selects equipment up to the point where the request is either accepted as Pending or rejected because the item is unavailable. This is one of the most suitable behaviors to model dynamically because it includes user actions, system validation, a decision branch, and both success and failure outcomes.

The diagram also demonstrates an important analysis decision: the system does not create a reservation first and validate later. Instead, validation is performed before the new reservation object is confirmed. That ordering is significant because it prevents inconsistent state, reduces avoidable conflicts, and directly supports the business rule that overlapping reservations are not permitted.

- The positive branch shows the normal path in which the equipment is available and the reservation iscreated with Pending status.
- The negative branch models abnormal behavior by returning a visible error path to the student rather thanallowing a silent failure.
- The partition between Student and System clarifies which actions belong to the user and which belong tothe software.

Figure 2. Activity Diagram

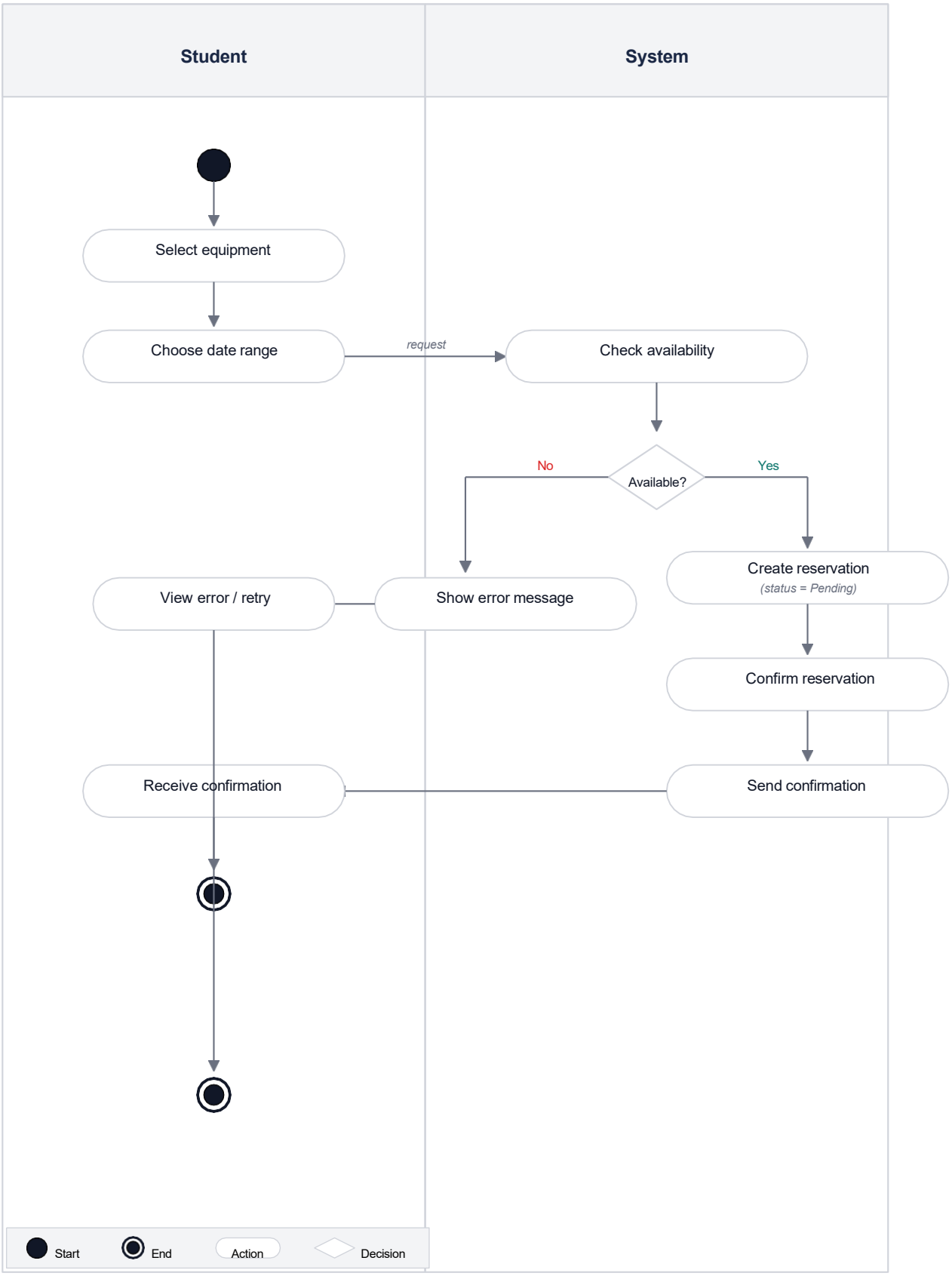


Figure 2. Activity Diagram for the Create Reservation use case.

11. UML Conceptual Class Diagram

The Conceptual Class Diagram captures the core domain concepts of UniGear Connect and shows how they relate at the analysis level. It intentionally avoids implementation-specific artifacts such as database technology,

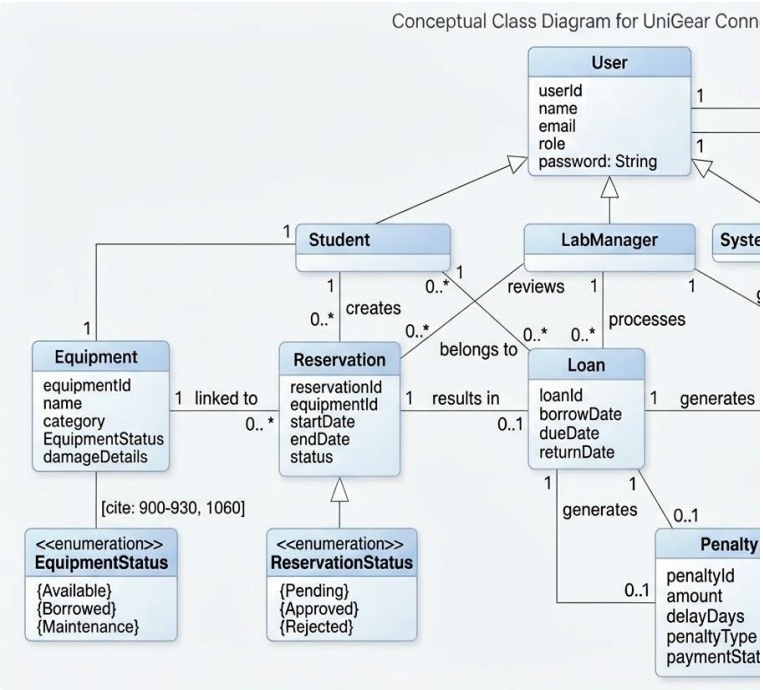
controllers, or service-layer classes. Instead, it focuses on the objects that matter in the real borrowing domain and on the associations that explain how borrowing actually works.

This diagram is especially important because it distinguishes between objects that are often incorrectly merged in weak models. For example, Reservation and Loan are not the same thing: a reservation expresses planned intent, while a loan represents actual possession after check-out. Similarly, Penalty is modeled as a consequence of an overdue loan rather than as a generic numeric attribute attached everywhere. This makes the domain logic clearer and improves future traceability into architecture and design.

- User is specialized into Student, LabManager, and SystemAdministrator so common identity data can beshared while responsibilities remain distinct.
- Equipment participates in many reservations and loans over time, but each operational loan concerns onespecific equipment item.
- Notifications and reports are modeled as supporting concepts because they communicate or summarizeevents derived from the operational workflow.

Figure 3. Conceptual Class Diagram

Figure 3. Conceptual Class Diagram for UniGear Connect.



12. Traceability: Functional Requirements to Use Cases

Traceability ensures that the requirements remain connected to observable system behavior. The following table maps each functional requirement to the use case or use cases that realize it.

Requirement	Mapped Use Case(s)
FR-01	UC-01 Search Equipment
FR-02	UC-01 Search Equipment
FR-03	UC-02 Create Reservation
FR-04	UC-02 Create Reservation
FR-05	UC-03 Review Reservation Request
FR-06	UC-04 Check-Out Equipment; UC-05 Return Equipment
FR-07	UC-01 Search Equipment; UC-02 Create Reservation
FR-08	UC-04 Check-Out Equipment; UC-05 Return Equipment
FR-09	UC-05 Return Equipment
FR-10	UC-05 Return Equipment
FR-11	UC-04 Check-Out Equipment; UC-05 Return Equipment; UC-06 Generate Overdue Report
FR-12	Administrative policy management (outside detailed UC set but covered by system administration scope)

13. Alignment: Use Cases to Domain Concepts

This second mapping shows how behavior aligns with conceptual analysis objects. It demonstrates that the use cases are grounded in the domain model rather than invented independently of it.

Use Case	Relevant Domain Concepts
UC-01 Search Equipment	Student, Equipment
UC-02 Create Reservation	Student, Equipment, Reservation, Notification, Borrowing Rule
UC-03 Review Reservation Request	LabManager, Reservation, Equipment, Notification
UC-04 Check-Out Equipment	LabManager, Reservation, Loan, Equipment, Student
UC-05 Return Equipment	LabManager, Loan, Equipment, Penalty

UC-06 Generate Overdue Report	LabManager, Loan, Equipment, Report, Student

14. Conclusion

UniGear Connect is a realistic educational resource management system with clear roles, policies, and state transitions. The requirements defined in this document are testable, bounded, and aligned with the earlier system brief. The conceptual model demonstrates an understanding of how reservation, loan, equipment status, penalties, notifications, and reports work together in the problem domain. This D2 deliverable provides a stable foundation for the later architecture and detailed design stages of the course project.